Additional case for GPK(Graham, Knuth and Patashnik) sequences

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Objective: To give the generating functions and study the asymptotic behavior of $T(n,k)$ defined by

$$T(n,k) = (a_2 n + a_1 k + a_0)\, T(n-1,k) + (b_2 n + b_1 k + b_0)\, T(n-1,k-1),$$

with the usual initial condition

$$T(0,0) = 1, \quad T(n,k) = 0 \text{ if } n < \max(k,0). \tag{1}$$

These sequences generalize many combinatorial numbers, such as Stirling numbers, Eulerian numbers, Lah numbers, and Whitney numbers. Graham, Knuth, and Patashnik posed this problem in their book [4]. Many years ago, Wilf [7] (and also Neuwirth [5]) found the generating function for the case $b_2 = 0$, which yields an explicit formula. Other cases remain open, both in terms of explicit formulas and asymptotics. For families generalizing Eulerian numbers, Hwang et al. [3] made a significant step using probabilistic methods to derive asymptotics in some cases as well. Hitczenko [8] did recently (2025) some cases using the generating function of $b_2 = 0$.

Here, we treat the case $a_2 = -a_1$ using context-grammar (Willian Chen sens) with Dumont tree-like structure associated (also system of differential equation combinatorial). We get the generating function and we try to give the asymptotic estimates and distribution of the random variables associated to $(T(n,k))$, using the basic theory in Flajolet [2].

I would like to emphasize the following This is not a piece of work intended for publication at this stage. It is simply part of my personal exercises on asymptotic methods that I studied earlier. Therefore, it has not undergone any formal verification, and I am not sure to what extent it contains new results or overlaps with existing work. It could potentially be combined with our paper, in preparation on context-free grammars (available on arXiv). This is a preliminary draft.

1 Generating functions for $a_1 + a_2 = 0$ (additional case)

We get these generating functions. The idea is to consider the grammar

$$\{x \to G(x) = (b_1+b_2)x^2+(a_1+a_2)xy; \quad y \to G(y) = b_2xy+a_2y^2; \quad z \to G(z) = (b_1+b_2+b_0)zx+(a_0+a_2)zy\} \tag{2}$$

and use the following provable fact. If $\mathcal{D} = G(x)\partial_x + G(y)\partial_y + G(z)\partial_z$ then

$$\mathcal{D}^n(z) = z\sum_{k=0}^{n} T(n,k)x^k y^{n-k}. \tag{3}$$

We apply this fact in the Dumont [1] (Randrianirina [9]) point of views. What we use here, can be summarized as the following.

Proposition 1.1. *Let $\vec{y}(t) = (y_i(t))_{i=1,\dots,v}$ be the solution of the analytic differential system*

$$y_i'(t) = G_i(\vec{y}(t)), \qquad y_i(0) = x_i, \quad i = 1,\dots,v. \tag{4}$$

Let $\mathcal{D} = \sum_{i=1}^{v} G(\vec{x}(t))\partial_{x_i}$. Then for each i, $\mathrm{Gen}(x_i,t) := \sum_{n\ge0} \mathcal{D}^n(x_i)\frac{t^n}{n!} = y_i(t)$.

On the other hand, the data of the grammar G, the differential operator D, and the differential system are equivalent. Here, we use only the analytic part of the differential system.
So firstly, we solve the system of differential equations associated with the grammar (2).

$$\begin{cases} X'(t) &= (b_1+b_2)X^2, \quad X(0)=x, \\ Y'(t) &= b_2XY + a_2Y^2, \quad Y(0)=y, \\ Z'(t) &= (b_0+b_2)ZX + (a_0+a_2)ZY, \quad Z(0)=z \end{cases} \tag{5}$$

Applying Proposition 1.1 with the fact (3), one can get

$$\sum_{n\geq 0} z \sum_{k=0}^{n} T(n,k)x^k y^{n-k}\frac{t^n}{n!} = \sum_{n\geq 0}\mathcal{D}^n(z)\frac{t^n}{n!} = Z(t,x,y,z).$$

We try to solve the system (5) and substitute $y=1$. So

$$F(t,x) = \frac{Z(t)}{z} = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!}. \tag{6}$$

....WILL WRITE THE CONCRETE (SHORT) WAY TO GET THE SOLUTION OF (5)....

In all cases, the generating functions can be derived explicitly because the system (5) can be solved using elementary functions. The explicit formulas are then obtained from these generating functions.

1.1 Case 1: $b_1 + b_2 = 0$

Case 1A: $a_2 \neq 0$, $b_2 = 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt)\,(1-a_2t)^{-\frac{a_0+a_2}{a_2}}.$$

Developing the generating function, we obtain

$$T(n,k) = \binom{n}{k} b_0^k\, a_2^{n-k}\, \left(\frac{a_0+a_2}{a_2}\right)_{n-k}.$$

Case 1B: $a_2 = 0$, $b_2 = 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt + a_0t)\,.$$

This is trivial, even without passing through generating functions. $T(n,k) = \binom{n}{k}a_0^k b_0^{n-k}$.

Case 1C: $a_2 \neq 0$, $b_2 \neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp\big(b_0x\,t\big)\left(\frac{b_2x}{b_2x+a_2-a_2\,e^{b_2xt}}\right)^{\frac{a_0+a_2}{a_2}}.$$

Expanding the generating function, we obtain

$$T(n,k) = \binom{-\frac{a_0+a_2}{a_2}}{n-k}\left(\frac{a_2}{b_2}\right)^{n-k}\sum_{j=0}^{n-k}(-1)^j\binom{n-k}{j}(b_0+jb_2)^n.$$

Case 1D: $a_2 = 0$, $b_2 \neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} T(n,k)x^k \frac{t^n}{n!} = \exp\left(b_0 xt + \frac{a_0}{b_2 x}\left(e^{b_2 xt} - 1\right)\right).$$

Developing the generating function, we obtain

$$T(n,k) = \frac{(a_0/b_2)^{n-k}}{(n-k)!}\sum_{j=0}^{n-k}(-1)^{n-k-j}\binom{n-k}{j}(b_0 + jb_2)^n.$$

We can have also

$$T(n,k) = \left(\frac{a_0}{b_2}\right)^{n-k}\sum_{r=n-k}^{n}\binom{n}{r} b_0^{n-r} b_2^r S(r, n-k),$$

$S(r,m)$ is Stirling numbers of the second kind.

1.2 Case 2: $b_1 + b_2 \neq 0$, $a_2 = 0$

Case 2A: $b_1 \neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} T(n,k)x^k \frac{t^n}{n!} = (1 - (b_1 + b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$
$$\times \exp\left[-\frac{a_0}{b_1 x}\left((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}} - 1\right)\right].$$

Expanding the generating function, we obtain

$$T(n,k) = \frac{n!}{(n-k)!}\left(-\frac{a_0}{b_1}\right)^{n-k}(-(b_1+b_2))^n \sum_{j=0}^{n-k}\binom{n-k}{j}(-1)^{(n-k)-j}\binom{\frac{b_1 j - (b_0+b_1+b_2)}{b_1+b_2}}{n}.$$

Case 2B: $b_1 = 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = (1 - b_2 xt)^{-\frac{(b_0+b_2)x + a_0}{b_2 x}}.$$

Developing the generating function, we obtain

$$T(n,k) = a_0^{n-k}\sum_{1\leq m_1 < \cdots < m_k \leq n}\prod_{r=1}^{k}(b_0 + m_r b_2),$$

or

$$T(n,k) = (-1)^{n-k}\left(\frac{a_0}{b_2}\right)^{n-k} b_2^n \sum_{m=n-k}^{n}\binom{n}{m} s(m,k)\,(-1)^m \left(\frac{b_0+b_2}{b_2}\right)_{n-m}.$$

1.3 Case 3: $b_1 + b_2 \neq 0$, $a_2 \neq 0$

Case 3A: $b_1 \neq 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}} \times \left(\frac{1}{1-\frac{a_2}{b_1 x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]} \right)^{\frac{a_0+a_2}{a_2}}.$$

Developing the generating function, we obtain

$$T(n,k) = \binom{\alpha+n-k-1}{n-k}\left(\frac{a_2}{b_1}\right)^{n-k} n!\,(b_1+b_2)^n \sum_{j=0}^{n-k}(-1)^j\binom{n-k}{j}\binom{\frac{b_0+b_1+b_2-jb_1}{b_1+b_2}+n-1}{n}.$$

Case 3B: $b_1 = 0$

We solve the system (5) for $Z(t)$, and then substitute into (6) to obtain

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = (1-b_2xt)^{-\frac{b_0+b_1}{b_2}-1} \times \left(\frac{1}{1+\frac{a_2}{b_2 x}\ln(1-b_2xt)} \right)^{\frac{a_0+a_2}{a_2}}.$$

Then we obtain

$$\sum_{n\geq 0} T(n,k)\frac{t^n}{n!} = (1-b_2t)^{-\frac{b_0+b_1}{b_2}-1}\cdot \frac{(-1)^k\left(\frac{a_0+a_2}{a_2}\right)_k}{k!}\left(\frac{a_2}{b_2}\ln(1-b_2t)\right)^k.$$

2 Analytic approach

For this case we assume that $(T(n,k)) \subset \mathbb{R}_{\geq 0}$.

As usual, we set the random variable X_n defined by

$$\mathbb{P}(X_n = k) := \frac{T(n,k)}{\sum_{j=0}^{n} T(n,j)}.$$

Considering the polynom

$$P_n(x) := \sum_{k=0}^{n} T(n,k)x^k = n![t^n]F(t,x),$$

we have then $F(t,x) = \sum_{n\geq 0} P_n(x)\frac{t^n}{n!}$ and X_n is the random variable of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{P_n(x)}{P_n(1)} = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}.$$

We use also tools from complex analysis to estimate an polymorphic function.

Definition 2.1 (Hayman–admissibility [2]). *Let $G(z)$ have radius of convergence ρ with $0 < \rho \leq +\infty$ and be always positive on some subinterval (R_0, ρ) of $(0, \rho)$. The function $G(z)$ is said to be admissible if it satisfies the following three conditions.*

- $\boldsymbol{H}_1$. *[Capture condition]* $\lim_{r \to \rho} b(r) = +\infty$.
- $\boldsymbol{H}_2$. *[Locality condition] For some function $\theta_0(r)$ defined over (R_0, ρ) and satisfying $0 < \theta_0 < \pi$, one has*
$$G(re^{i\theta}) \sim G(r) e^{i\theta a(r) - \theta^2 b(r)/2} \quad \textit{as } r \to \rho,$$
uniformly in $|\theta| \leq \theta_0(r)$.
- $\boldsymbol{H}_3$. *[Decay condition] Uniformly in $\theta_0(r) \leq |\theta| < \pi$,*
$$G(re^{i\theta}) = o\left(\frac{G(r)}{\sqrt{b(r)}}\right).$$

Here

$$a(r) = r\frac{G'(r)}{G(r)} = r(\log G(r))', \quad b(r) = r\frac{G'(r)}{G(r)} + r^2 \frac{G''(r)}{G(r)} - r^2 \left(\frac{G'(r)}{G(r)}\right)^2 = r(\log G(r))' + r^2(\log G(r))''.$$

Theorem 2.1 (Coefficients of admissible functions [2]). *Let $G(z)$ be an H–admissible function and $\zeta \equiv \zeta(n)$ be the unique solution in the interval (R_0, ρ) of the saddle point equation*

$$\zeta \frac{G'(\zeta)}{G(\zeta)} = n.$$

The Taylor coefficients of $G(z)$ satisfy, as $n \to \infty$

$$g_n \equiv [z^n]G(z) \sim \frac{G(\zeta)}{\zeta^n \sqrt{2\pi b(\zeta)}} \quad \textit{as } n \to \infty$$

with $b(z) = z^2 h''(z) + z h'(z)$ and $h(z) = \log G(z)$.

Theorem 2.2 (Generalized quasi-powers [2]). *Assume that, for u in a fixed neighborhood Ω of 1, the generating function $p_n(u)$ of a non-negative discrete random variable X_n admits a representation of the form*

$$p_n(u) = \exp\big(h_n(u)\big)\,(1 + o(1)),$$

that holds uniformly, where each $h_n(u)$ is analytic in Ω. Assume also the condition,

$$h'_n(1) + h''_n(1) \longrightarrow +\infty \qquad \textit{and} \qquad \frac{h'''_n(u)}{(h'_n(1) + h''_n(1))^{3/2}} \longrightarrow 0,$$

uniformly for $u \in \Omega$. Then, the random variable

$$X_n^* = \frac{X_n - h'_n(1)}{(h'_n(1) + h''_n(1))^{1/2}}$$

converges in distribution to a normal law with parameters $(0, 1)$.

Theorem 2.3 (Algebraic singularity schema [2]). *Let $F(z, u)$ be a function that is bivariate analytic at $(z, u) = (0, 0)$ with non-negative coefficients. Assume the following conditions:*

(i) ***Analytic perturbation:*** *there exist three functions A, B, C, analytic in a domain $\mathcal{D} = \{|z| \le r\} \times \{|u-1| < \epsilon\}$, such that for some r_0 with $0 < r_0 < r$, and $\epsilon > 0$, the following representation holds, with $\alpha \notin \mathbb{Z}_{\le 0}$,*

$$F(z,u) = A(z,u) + B(z,u)C(z,u)^{-\alpha};$$

furthermore, assume that, in $|z| \le r$, there exists a unique root ρ of the equation $C(z,1) = 0$ that this root is simple and that $B(\rho, 1) \neq 0$.

(ii) ***Nondegeneracy:*** *one has $\partial_z C(\rho,1) \cdot \partial_u C(\rho,1) \neq 0.$, ensuring the existence of a non-constant $\rho(u)$, analytic at $u = 1$, such that $C(\rho(u), u) = 0$ and $\rho(1) = \rho$.*

(iii) ***Variability:*** *one has*

$$\mathfrak{v}\big(\frac{\rho}{\rho(x)}\big) = -\frac{\rho''(1)}{\rho(1)} - \frac{\rho'(1)}{\rho(1)} + \left(\frac{\rho'(1)}{\rho(1)}\right)^2 \neq 0.$$

Then, the random variable with probability generating function

$$p_n(u) = \frac{[z^n]F(z,u)}{[z^n]F(z,1)}$$

converges to a Gaussian law with speed $O(n^{-1/2})$.

In this case $V(X_n) \sim \sigma_n^2 \sim \mathfrak{v}\big(\frac{\rho}{\rho(x)}\big)n$ and $E(X_n) \sim \mu_n \sim \mathfrak{m}\big(\frac{\rho(1)}{\rho(u)}\big)n$, for $\mathfrak{m}\big(\frac{\rho(1)}{\rho(u)}\big) = -\frac{\rho'(1)}{\rho(1)}$.

2.1 Case 1A $a_1 = -a_2 \neq 0$, $b_1 = b_2 = 0$

From

$$F(t,x) = \sum_{n\ge 0}\sum_{k=0}^{n} T(n,k)x^k \frac{t^n}{n!} = \exp(b_0 x t)\,(1-a_2 t)^{-\frac{a_0+a_2}{a_2}},$$

we get

$$\sum_{n\ge 0} T(n,k)\frac{t^n}{n!} = \frac{(b_0 t)^k}{k!}(1-a_2 t)^{-\frac{a_0+a_2}{a_2}}.$$

Then

$$T(n,k) = \frac{n! b_0^k}{k!}[t^n]t^k(1-a_2 t)^{-\frac{a_0+a_2}{a_2}} = \frac{n! b_0^k}{k!}[t^{n-k}](1-a_2 t)^{-\frac{a_0+a_2}{a_2}}.$$

So if k is not near to n ($n-k$ is huge) and $n \to \infty$

$$T(n,k) \sim \frac{b_0^k}{k!\,\Gamma\left(\frac{a_0+a_2}{a_2}\right)}\, a_2^{n-k}\,(n-k)^{\frac{a_0}{a_2}}\,\sqrt{2\pi n}\left(\frac{n}{e}\right)^n. \tag{7}$$

We note that the asymptotic at (7) is sure only for $\frac{a_0+a_2}{a_2} \notin \mathbb{Z}_{\le 0}$. Otherwise, a_2 divide a_0, opposite sign, let say $a_0 = -q a_0$, for $q \in \mathbb{N}$. So $(1-a_2 t)^{-\frac{a_0+a_2}{a_2}} = (1-a_2 t)^{q-1}$ is of degree $q-1$. Then $T(n,k) = 0$ unless $k \ge n+1-q$, where $T(n,k) = \frac{n! b_0^k}{k!}(-a_2^{n-k})\binom{\frac{a_0+a_2}{a_2}}{n-k}$.

Proposition 2.1. *Let $(T(n,k))$ be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k) = (a_2 n - a_2 k + a_0)\,T(n-1,k) + b_0\,T(n-1,k-1).$$

Assume $\frac{a_0+a_2}{a_2} \notin \mathbb{Z}_{\le 0}$ and all coefficients are non-negative. $F(t,x)$ be the exponential bivariate generating function. Then The random variable X_n of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a Poisson random variable.

Proof. We write the asymptotic of $p_n(x)$. First, we know

$$[t^n](1-a_2t)^{-\alpha} = \frac{a_2^n}{\Gamma(\alpha)}\, n^{\alpha-1}\left(1+O\left(\frac{1}{n}\right)\right).$$

Also e^{b_0xt} is analytic at the singularity $\rho = \frac{1}{a_2}$, then

$$[t^n]F(t,x) = e^{b_0x/a_2}\frac{a_2^n}{\Gamma(\frac{a_0+a_2}{a_2})}\, n^{\frac{a_0}{a_2}}\left(1+O\left(\frac{1}{n}\right)\right)$$

and

$$p_n(x) = \frac{e^{b_0x/a_2}}{e^{b_0/a_2}}\cdot\frac{1+O(1/n)}{1+O(1/n)}.$$

So we have

$$p_n(x) \longrightarrow \exp(\lambda(x-1)), \qquad \lambda = \frac{b_0}{a_2}.$$

□

2.2 Case 1B $a_1 = a_2 = 0$, $b_1 = b_2 = 0$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt + a_0t).$$

This case is trivial, in many different ways, we can get quickly that

$$T(n,k) = \binom{n}{k}a_0^{n-k}b_0^k \qquad \Longrightarrow\ X_n \sim \mathrm{Bin}(n, \frac{b_0}{(a_0+b_0)}).$$

2.3 Case 1C: $a_1 = -a_2 \neq 0$, $b1 = -b_2 \neq 0$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp\big(b_0x\,t\big)\left(\frac{b_2x}{b_2x + a_2 - a_2\,e^{b_2xt}}\right)^{\frac{a_0+a_2}{a_2}}.$$

2.4 Case 1D: $a_2 = 0$, $b_1 = -b_2 \neq 0$

$$T(n,k) = a_0\,T(n-1,k) + (b_2n - b_2k + b_0)\,T(n-1,k-1).$$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = \exp\left(b_0xt + \frac{a_0}{b_2x}\big(e^{b_2xt}-1\big)\right). \tag{8}$$

Equation (8) is equal to

$$\sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)(\frac{1}{x})^{n-k}\frac{(xt)^n}{n!} = \exp\left(b_0xt + \frac{a_0}{b_2x}\big(e^{b_2xt}-1\big)\right)$$

$$\Longrightarrow \sum_{n\geq 0}\sum_{k=0}^{n} T(n,n-k)u^k\frac{z^n}{n!} = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)u^{n-k}\frac{z^n}{n!} = e^{b_0z}\exp\left(\frac{a_0}{b_2}\big(e^{b_2z}-1\big)u\right)$$

$$\Longrightarrow \sum_{n\geq 0} T(n,n-k)\frac{z^n}{n!} = \frac{a_0^k}{k!b_2^k}e^{b_0z}\big(e^{b_2z}-1\big)^k. \tag{9}$$

So

$$T(n,k) = (\frac{n!a_0^k}{k!b_2^k})[z^n]G_k(z), \qquad G_k(z) = e^{b_0z}\big(e^{b_2z}-1\big)^k. \tag{10}$$

Let's check quickly the admissibility of $G_k(z)$. It is always positive on $(R_0 = 0, \rho = +\infty)$. We compute first $a(r)$ and $b(r)$ and get

$$a(r) = r\left(b_0 + k\frac{b_2 e^{b_2 r}}{e^{b_2 r} - 1}\right), \quad b(r) = r\left(b_0 + k\frac{b_2 e^{b_2 r}}{e^{b_2 r} - 1}\right) - r^2 k\frac{b_2^2 e^{b_2 r}}{(e^{b_2 r} - 1)^2}.$$

Then for $b_0, b_2 > 0$ H1, capture condition is verified: $\lim_{n\to+\infty} b(r) = +\infty$. We expend $e^{b_0 r e^{i\theta}}$ and $(e^{b_1 r e^{i\theta}} - 1)$ wisely around $\theta = 0$ (small enough θ) to get H2, locality condition

$$G(re^{i\theta}) \sim G(r)e^{i\theta a(r) - \theta^2 b(r)/2} \quad \text{as } r \to +\infty.$$

Let $c = (b_0 + kb_2)(1 - \cos(\theta_0))$ so that for $\theta_0(r) \le \theta \le \pi$, $(b_0 + kb_2)(\cos\theta - 1) \le -c$ and

$$\left|\frac{G(re^{i\theta})}{G(r)/\sqrt{b(r)}}\right| = \sqrt{b(r)}\, e^{(b_0+kb_2)r(\cos\theta-1)}\,(1 + o(1)) \le \sqrt{b(r)}\, e^{-cr} \longrightarrow 0, \quad (r \to \infty,\ |\theta| \ge \theta_0(r)).$$

So H3, decay condition is satisfied, thus G_k is H-admissible. The saddle point equation is given by

$$\zeta_k(n)\left(b_0 + k\frac{b_2 e^{b_2\zeta_k(n)}}{e^{b_2\zeta_k(n)} - 1}\right) = n. \tag{11}$$

We will apply the Coefficients admissible theorem. Let's estimate first $\zeta_k(n)$

$$\zeta_k(n)\left(\frac{b_0}{n} + \frac{k}{n}\frac{b_2 e^{b_2\zeta(n)}}{e^{b_2\zeta(n)} - 1}\right) = 1.$$

So if $n \to \infty$ (we assume that b_0 is not very high comparing to b_2), we have

$$\zeta_k(n)\frac{k}{n}\frac{b_2 e^{b_2\zeta_k(n)}}{e^{b_2\zeta_k(n)} - 1} \sim 1 \quad \text{or} \quad \frac{b_2\zeta_k(n)}{1 - e^{-b_2\zeta_k(n)}} \sim \frac{n}{k}.$$

Therefore $\zeta_0(n) = n/b_0$ and for $k \neq 0$

$$\zeta_k(n) \sim \frac{1}{b_2}\left(\frac{n}{k} + W\Big(-\frac{n}{k}e^{-\frac{n}{k}}\Big)\right). \tag{12}$$

Thus the Coefficients admissible theorem gives us the following results.

Proposition 2.2. *Let $(T(n,k))$ be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k) = a_0\,T(n-1,k) + (b_2 n - b2k + b_0)\,T(n-1,k-1).$$

We have

$$T(n,n-k) \sim \Big(\frac{n!a_0^k}{k!b_2^k}\Big)\frac{G_k(\zeta_k(n))}{\zeta_k(n)^n\sqrt{2\pi n\big(1 - \frac{b_2\zeta_k(n)}{(e^{b2\zeta_k(n)}-1)}\big)}}.$$

Equivalently

$$T(n,k) \sim \Big(\frac{n!a_0^{n-k}}{(n-k)!b_2^{n-k}}\Big)\frac{e^{b_0\zeta_{n-k}(n)}\big(e^{b_2\zeta_{n-k}(n)} - 1\big)^{n-k}}{\zeta_{n-k}(n)^n\sqrt{2\pi n\big(1 - \frac{b_2\zeta_{n-k}(n)}{(e^{b2\zeta_{n-k}(n)}-1)}\big)}},$$

for $\zeta_{n-k}(n)$ given by (11) *that can be as in* (12) *or $\zeta_{n-k}(n) \sim \frac{n}{b_0+(n-k)b_2}$, accordingly ($k \ll n$ or $n - k \ll n$) or not.*

Here $k \ll n$ is also depending on how big $b_0 \gg b_2$. Somehow, $k \ll n$ should be $kf(b_0) \ll ng(b_2)$ for some increasing functions f and g.

(n,k)	Exact value	Approximate value	Relative error
$(20000, 5)$	$1.245852907 \times 10^{18100}$	$1.280350540 \times 10^{18100}$	0.0276906
$(20000, 10)$	$9.735206503 \times 10^{18134}$	$9.732543012 \times 10^{18134}$	0.0002736
$(20000, 50)$	$1.001227349 \times 10^{18392}$	$1.003427242 \times 10^{18392}$	0.0021962
$(20000, 100)$	$1.127882259 \times 10^{18692}$	$1.129909884 \times 10^{18692}$	0.0017976
$(20000, 300)$	$1.697672189 \times 10^{19808}$	$1.698279109 \times 10^{19808}$	0.0003574
$(20000, 700)$	$4.043920535 \times 10^{21871}$	$4.044741395 \times 10^{21871}$	0.0002030
$(20000, 1000)$	$5.606744636 \times 10^{23340}$	$5.607414282 \times 10^{23340}$	0.0001194
$(20000, 1200)$	$2.386834815 \times 10^{24294}$	$2.387338510 \times 10^{24294}$	0.0002110
$(20000, 1500)$	$3.426108298 \times 10^{25693}$	$3.426491193 \times 10^{25693}$	0.0001117
$(20000, 1900)$	$1.334384744 \times 10^{27510}$	$1.334538680 \times 10^{27510}$	0.0001153
$(20000, 1950)$	$5.227751444 \times 10^{27733}$	$5.228480286 \times 10^{27733}$	0.0001394
$(20000, 1990)$	$1.182081908 \times 10^{27912}$	$1.182215891 \times 10^{27912}$	0.0001134
$(20000, 1995)$	$2.161821454 \times 10^{27934}$	$2.162018694 \times 10^{27934}$	0.0000913

Table 1: Comparison of exact and approximate values with relative errors for $a_0 = 8, b_0 = 4, b_2 = 3$.

On another hand, from the Cauchy Integral formula, the equation (9) implies

$$T(n,k) = \frac{n!}{2\pi i} \frac{a_0^{n-k}}{(n-k)! b_2^{n-k}} \int_{\mathcal{C}(0,\rho)} \frac{e^{b_0 z}\left(e^{b_2 z} - 1\right)^{n-k}}{z^{n+1}} \, dz. \tag{13}$$

One can do is, use similar thing what Corcino [10] estimate the integral, for $r - Whitney$ numbers.

Theorem 2.4. *Let $(T(n,k))$ be sequence satisfying the usual initial condition (1) and*

$$T(n,k) = a_0\, T(n-1,k) + (b_2 n - b2k + b_0)\, T(n-1,k-1).$$

Let $\rho^* > 0$ be the unique solution of the saddle-point equation

$$\rho^* \left(b_0 + \frac{(n-k)b_2}{1 - e^{-b_2\rho^*}} \right) = n. \tag{14}$$

Define

$$B = \frac{1}{2}\left[\frac{b_0}{n-k} + \frac{b_2 e^{b_2\rho^*}(e^{b_2\rho^*} - b_2\rho^* - 1)}{(e^{b_2\rho^*} - 1)^2} \right]. \tag{15}$$

Then, for $\frac{b_0}{b2} \geq 4/3$,

$$T(n,k) \sim \frac{n!}{(n-k)!} \frac{a_0^{n-k}}{b_2^{n-k}} \frac{e^{b_0\rho^*}(e^{b_2\rho^*} - 1)^{n-k}}{2\,(\rho^*)^n \sqrt{\pi (n-k)\rho^* B}} (1+s), \tag{16}$$

with

$$s=\frac{1}{(n-k)\,\rho^*}\Bigg\{\frac{3}{4\rho^*B^2}\Bigg[\frac{b_0\rho^*}{24(n-k)}+\frac{1}{24}\Big(b_2\rho^*+(b_2\rho^*-7(b_2\rho^*)^2+6(b_2\rho^*)^3-(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-1}\Big)$$
$$+\frac{1}{24}\Big((18(b_2\rho^*)^3-7(b_2\rho^*)^2-7(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-2}\Big)$$
$$+(12(b_2\rho^*)^3-12(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-3}-\frac{1}{24}\Big(6(b_2\rho^*)^4(e^{b_2\rho^*}-1)^{-4}\Big)\Bigg]$$
$$-\frac{15}{16(\rho^*)^2B^3}\Bigg[\frac{b_0\rho^*}{6(n-k)}+\frac{1}{6}\Big(b_2\rho^*+(b_2\rho^*-3(b_2\rho^*)^2+(b_2\rho^*)^3)(e^{b_2\rho^*}-1)^{-1}\Big)$$
$$+(3(b_2\rho^*)^3-3(b_2\rho^*)^2)(e^{b_2\rho^*}-1)^{-2}+\frac{1}{6}\Big(2(b_2\rho^*)^3(e^{b_2\rho^*}-1)^{-3}\Big)\Bigg]^2\Bigg\}.$$

Also, it is still true even $q:=\frac{b_0}{b2}<4/3$, *if* $k\leq n/q$.

We note that the ρ^* at (14) is nothing else than $\zeta_{n-k}(n)$, according to (11).

Proof. From (13),

$$T(n,k)=\frac{n!}{2\pi i}\frac{a_0^{n-k}}{(n-k)!b_2^{n-k}}I,\quad I=\int_{\mathcal{C}(0,R)}\frac{e^{b_0z}\big(e^{b_2z}-1\big)^{n-k}}{z^{n+1}}\,dz. \tag{17}$$

We parametrize the contour by

$$z=\rho e^{i\theta},\qquad dz=i\rho e^{i\theta}d\theta.$$

Substituting into the integral gives

$$I=\int_{-\pi}^{\pi}\frac{e^{b_0\rho e^{i\theta}}\big(e^{b_2\rho e^{i\theta}}-1\big)^{n-k}}{(\rho e^{i\theta})^{n+1}}\,i\rho e^{i\theta}\,d\theta=i\rho^{-n}\int_{-\pi}^{\pi}\exp\big(b_0\rho e^{i\theta}\big)\big(e^{b_2\rho e^{i\theta}}-1\big)^{n-k}e^{-in\theta}\,d\theta.$$

Thus,

$$I=i\rho^{-n}\int_{-\pi}^{\pi}\exp(h_\rho(\theta))\,d\theta,\quad h_\rho(\theta)=b_0\rho e^{i\theta}+(n-k)\,\log\big(e^{b_2\rho e^{i\theta}}-1\big)-in\theta.$$

The next step is expand $h_\rho(\theta)$ and choose ρ^*, saddle point (at $\theta=0$): we choose is as radius of the countour. So in the expansion, the order zero is constant (can put out of the integral); the order is zero by definition.We have then

$$T(n,k)=(\textbf{constant})\times\int_{-\pi}^{\pi}\exp\big(-(n-k)B\theta^2\big)\exp\big(\textbf{oder superior to 3}\big).$$

Now, we want to bring it to gamma function, so we need $x^2=(n-k)B\theta^2$.
The next task is to argue one the domain of integral, so it can extend to infinity and many computation to have the result. For many details of these step, please see [11],. □

Proposition 2.3. *Let* $(T(n,k))$ *be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k)=a_0\,T(n-1,k)+(b_2n-b2k+b_0)\,T(n-1,k-1).$$

Assume that all coefficients are non-negative. $F(t,x)$ *be the exponential bivariate generating function. Then The random variable* X_n *of probability generating function*

$$p_n(x)=\mathbb{E}(x^{X_n})=\frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a a Gaussian with mean and variance

$$\mu_n \sim n - \frac{n}{W(\frac{b_2 n}{a_0})} + \frac{a_0}{b_2}; \qquad \sigma_n^2 \sim \frac{n}{W(\frac{b_2 n}{a_0})\big(1+W(\frac{b_2 n}{a_0})\big)} - \frac{a_0}{b_2}.$$

Proof. We notice first that the non-negativity of all $T(n,k)$ implies a_0, b_0 and b_2 are non-negative.

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = \exp\left(b_0 x t + \frac{a_0}{b_2 x}\left(e^{b_2 x t} - 1\right)\right). \tag{18}$$

We can also see that $F_x(t) = F(t,x)$ is H-admissible for $x > 0$ with

$$\rho = +\infty, \quad a(r) = r\left(b_0 x + a_0 e^{b_2 x r}\right), \quad b(r) \ = a_0 b_2 x\, r^2 e^{b_2 x r} + a_0 r e^{b_2 x r} + b_0 x\, r.$$

Then Coefficients of admissible functions theorem (Theorem 2.1) ensures us

$$[t^n]F(t,x) \sim \frac{\exp\Big(b_0 x \zeta_x(n) + \frac{a_0}{b_2 x}\left(e^{b_2 x \zeta_x(n)} - 1\right)\Big)}{\zeta_x(n)^n \sqrt{2\pi\left(a_0 b_2 x\, \zeta_x(n)^2 e^{b_2 x \zeta_x(n)} + n\right)}} \tag{19}$$

where $\zeta_x(n)\left(b_0 x + a_0 e^{b_2 x \zeta_x(n)}\right) = n$.
We will use the Generalized quasi-Powers theorem. We need to write $p_n(x) = \exp(h_n(t,x))$. Let's start by approximating $p_n(x)$ near $x = 1$. If $h(t,x) = b_0 x t + \frac{a_0}{b_2 x}\left(e^{b_2 x t} - 1\right)$ then $F(t,x) = \exp(h(t,x))$ and the saddle point equation is

$$\frac{\partial}{\partial t} h(t,x)\Big|_{t=\zeta_x(n)} = \frac{n}{\zeta_x(n)} \implies \zeta_x(n)\Big(b_0 x + a_0 e^{b_2 x \zeta_x(n)}\Big) = n. \tag{20}$$

We get now (19) and compute

$$p_n(x) \sim \frac{F(\zeta_x(n), x)}{F(\zeta_1(n), 1)} \left(\frac{\zeta_1(n)}{\zeta_x(n)}\right)^n K_n(x)(1 + o(1)), \quad K_n(x) = \sqrt{\frac{n + a_0 b_2 \zeta_1(n)^2 e^{b_2 \zeta_1(n)}}{n + a_0 b_2 x \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}}}.$$

The $h_n(x)$ wanted is then

$$h_n(x) = \log p_n(x) = h(\zeta_x(n), x) - h(\zeta_1(n), 1) + n \log\left(\frac{\zeta_1(n)}{\zeta_x(n)}\right) + \log K(x) + o(1).$$

Using elementary calculus, chain rule with equation (20),

$$h_n'(x) = \partial_x h(\zeta_x(n), x) + K_n'(x)/K_n(x). \tag{21}$$

Then

$$h_n'(1) \sim b_0 \zeta_1(n) + a_0 \zeta_1(n) e^{b_2 \zeta_1(n)} - \frac{a_0}{b_2}(e^{b_2 \zeta_1(n)} - 1) + K_n'(1)/K_n(1). \tag{22}$$

If $D(x) = n + a_0 b_2 x \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}$, $K_n'(x)/K_n(x) = -\frac{1}{2}\frac{D'(x)}{D(x)}$. Then

$$\frac{K'(x)}{K(x)} = -\frac{1}{2} \frac{a_0 b_2 e^{b_2 x \zeta_x(n)} \Big(\zeta_x(n)^2 + 2x\, \zeta_x(n)\, \partial_x \zeta_x(n) + b_2 x\, \zeta_x(n)^3 + b_2 x^2\, \zeta_x(n)^2\, \partial_x \zeta_x(n)\Big)}{n + a_0 b_2 x\, \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}},$$

with

$$\partial_x \zeta_x(n) = -\frac{b_0\, \zeta_x(n) + a_0 b_2\, \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}}{b_0 x + a_0 e^{b_2 x \zeta_x(n)} + a_0 b_2 x\, \zeta_x(n) e^{b_2 x \zeta_x(n)}}.$$

So,

$$\frac{K'(x)}{K(x)} = -\frac{1}{2}\frac{a_0 b_2 e^{b_2 x\zeta_x(n)}}{n + a_0 b_2 x\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}\Bigg[\zeta_x(n)^2 + b_2 x\,\zeta_x(n)^3 - \left(2x\,\zeta_x(n) + b_2 x^2\,\zeta_x(n)^2\right)\frac{b_0\,\zeta_x(n) + a_0 b_2\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}{b_0 x + a_0 e^{b_2 x\zeta_x(n)} + a_0 b_2 x\,\zeta_x(n) e^{b_2 x\zeta_x(n)}}\Bigg].$$

As asymptotically, we have $b_0 x + a_0 e^{b_2\zeta_x(n)} \sim a_0 e^{b_2\zeta_x(n)}$, the saddle point equation gives us

$$\zeta_x(n) a_0 e^{b_2\zeta_x(n)} \sim n.$$

First, the prefactor is $\frac{a_0 b_2 e^{b_2 x\zeta_x(n)}}{n + a_0 b_2 x\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}} \sim \frac{1}{\zeta_x(n) + x\zeta_x(n)^2} \sim \frac{1}{x\zeta_x(n)^2}$. For the fraction in bracket, exponential terms dominate: $\frac{b_0\zeta_x(n) + a_0 b_2\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}{b_0 x + a_0 e^{b_2 x\zeta_x(n)} + a_0 b_2 x\zeta_x(n) e^{b_2 x\zeta_x(n)}} \sim \frac{\zeta_x(n)}{x}$. So the bracket: $\zeta_x(n)^2 + b_2 x\zeta_x(n)^3 - (2x\zeta_x(n) + b_2 x^2\zeta_x(n)^2)\frac{\zeta_x(n)}{x} = -\zeta_x(n)^2$. Finally, $\frac{K'(x)}{K(x)} \sim -\frac{1}{2}\cdot\frac{1}{x\zeta_x(n)^2}\cdot(-\zeta_x(n)^2) = \frac{1}{2x}$. Therefore (22) implies

$$h_n'(1) \sim n - \frac{a_0}{b_2} e^{b_2\zeta_1(n)}. \tag{23}$$

To estimate $\frac{a_0}{b_2} e^{b_2\zeta_1(n)}$, let $y_n = \zeta_1(n)$. Then asymptotically (20) becomes

$$y_n a_0 e^{b_2 y_n} \sim n. \tag{24}$$

Taking logarithm both sides, we have $\log(y_n) + \log(a_0) + b_2 y_n \sim \log(n)$. Noting that y_n go to infinity as $n \to \infty$, $b_2 y_n \sim \log(y_n) + \log(a_0) + b_2 y_n \sim \log(n)$. Then

$$y_n \sim \frac{\log(n)}{b_2}. \tag{25}$$

So (24) implies $\frac{a_0}{b_2} e^{b_2 y_n} \sim \frac{1}{b_2} \times \frac{n}{y_n} \sim \frac{n}{\log(n)}$. Thus (23) gives us

$$h_n'(1) \sim n - \frac{n}{\log(n)}.$$

If we want more precision than (25), the equation (24) can give us $b_2 y_n e^{b_2 y_n} \sim \frac{b_2 n}{a_0}$, that means $b_2 y_n = W(\frac{b_2 n}{a_0})$. So $y_n \sim \frac{W(\frac{b_2 n}{a_0})}{b_2}$. With (24) again, $\frac{a_0}{b_2} e^{b_2 y_n} \sim \frac{1}{b_2} \times \frac{n}{y_n} \sim \frac{n}{W(\frac{b_2 n}{a_0})}$. Thus (23) gives us

$$h_n'(1) \sim n - \frac{n}{W(\frac{b_2 n}{a_0})} + \frac{a_0}{b_2}. \tag{26}$$

For the same reasoning as we did above for $\frac{K'(x)}{K(x)}$, we got also similar thing for its derivative. So from (21)

$$h_n''(x) \sim \partial_x\zeta_x(n) h_t(\zeta_x(n), x) + h_x x(\zeta_x(n), x).$$

After computation, we get

$$h_n''(1) = \frac{2a_0}{b_2}\left(e^{b_2 y_n} - 1\right) - 2a_0 y_n e^{b_2 y_n} + a_0 b_2 y_n^2 e^{b_2 y_n} - \frac{y_n^2\left(b_0 + a_0 b_2 y_n e^{b_2 y_n}\right)^2}{n + a_0 b_2 y_n^2 e^{b_2 y_n}}.$$

Then

$$\begin{aligned} h_n'(1) + h_n''(1) &\sim n + \frac{a_0}{b_2}\left(e^{b_2 y_n} - 1\right) - 2n + nW(\frac{b_2 n}{a_0}) - \frac{\left(nW(\frac{b_2 n}{a_0})\right)^2}{n\left(1 + W(\frac{b_2 n}{a_0})\right)} \\ &\sim -n + \frac{n}{W(\frac{b_2 n}{a_0})} + nW(\frac{b_2 n}{a_0}) - \frac{nW(\frac{b_2 n}{a_0})^2}{\left(1 + W(\frac{b_2 n}{a_0})\right)} - \frac{a_0}{b_2} \\ &\sim \frac{n}{W(\frac{b_2 n}{a_0})\left(1 + W(\frac{b_2 n}{a_0})\right)} - \frac{a_0}{b_2}. \end{aligned} \tag{27}$$

After computation

$$
\begin{aligned}
h_n'''(x) &= -\frac{6a_0}{b_2x^4}\left(e^{b_2x\zeta_x(n)}-1\right)\\
&\quad+\frac{6a_0\zeta_x(n)}{x^3}e^{b_2x\zeta_x(n)}-\frac{3a_0b_2\zeta_x(n)^2}{x^2}e^{b_2x\zeta_x(n)}+\frac{a_0b_2^2\zeta_x(n)^3}{x}e^{b_2x\zeta_x(n)}+2a_0b_2^2\zeta_x(n)^2e^{b_2x\zeta_x(n)}\,\partial_x\zeta_x(n)\\
&\quad+a_0b_2e^{b_2x\zeta_x(n)}\Big(1+b_2x\zeta_x(n)\Big)\big(\partial_x\zeta_x(n)\big)^2+\Big(b_0+a_0b_2\zeta_x(n)e^{b_2x\zeta_x(n)}\Big)\partial_{xx}\zeta_x(n)+\left(\frac{K'}{K}\right)''(x).
\end{aligned}
$$

Noticing $\zeta_x(n)\sim\dfrac{\log n}{b_2x}$, $\partial_x\zeta_x(n)\sim-\dfrac{\log n}{b_2x^2}$ and $\partial_{xx}\zeta_x(n)\sim\dfrac{2\log n}{b_2x^3}$, we get asymptotically $h_n'''(x)\sim\frac{2n\log(n)}{x^3}$ and

$$
\frac{h_n'''(x)}{\big(h_n'(1)+h_n''(1)\big)^{3/2}}\sim\frac{2}{x^3}\frac{(\log(n))^4}{\sqrt{n}}\to 0, \tag{28}
$$

uniformly for $|x-1|<\frac{1}{2}$. Moreover, from (27) $h_n'(1)+h_n''(1)\longrightarrow+\infty$. So all of the assumption of the generalized quasi-powers are satisfied. Then Theorem 2.2 allow us to conclude that the associated X_n converges in distribution to a normal law with mean $h_n'(1)$ and variance $h_n'(1)+h_n''(1)$. □

n	$E(X_n)$	μ_n	Relative error	$V(X_n)$	σ_n^2	Relative error
50	33.062789	31.746100	0.03982	4.168800	4.042883	0.03020
100	69.971810	68.568867	0.02005	6.704993	6.601507	0.01543
200	146.874270	145.402395	0.01002	10.812412	10.727090	0.00789
500	386.556357	385.012809	0.00399	20.511749	20.444852	0.00326
1000	797.310132	795.723215	0.00199	33.605167	33.548907	0.00167
1200	963.621814	962.024722	0.00166	38.324153	38.270311	0.00140
1500	1214.494003	1212.885068	0.00132	45.052296	45.001228	0.00113
1900	1550.995102	1549.374302	0.00105	53.532933	53.484596	0.00090
2000	1635.411735	1633.788447	0.00099	55.582951	55.535180	0.00086
2500	2058.911485	2057.277717	0.00079	65.497272	65.451862	0.00069

Table 2: Comparison of exact and approximate values for $a_0=8, b_0=10, b_2=5$.

2.5 Case2A: $b_1+b_2\neq 0$, $a_2=0$ and $b_1\neq 0$ $(b_1<0)$

$$
\begin{aligned}
F(t,x)=\sum_{n\geq 0}\sum_{k=0}^{n}T(n,k)x^k\frac{t^n}{n!} &= (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}\\
&\quad\times\exp\left[-\frac{a_0}{b_1x}\left((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}-1\right)\right].
\end{aligned}
$$

Note that the singularity at $x=0$ is removable.
It is convenient to write $F(t,x)=\exp(h(t,x))$, where

$$
h(t,x)=-\frac{b_0+b_1+b_2}{b_1+b_2}\log\big(1-(b_1+b_2)xt\big)-\frac{a_0}{b_1x}\Big((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}-1\Big).
$$

The saddle point $\zeta_x(n)$ is defined implicitly by

$$
\frac{\partial}{\partial t}h(t,x)\Big|_{t=\zeta_x(n)}=\frac{n}{\zeta_x(n)}.
$$

So
$$\frac{(b_0+b_1+b_2)x}{1-(b_1+b_2)x\zeta_x(n)} + a_0\left(1-(b_1+b_2)x\zeta_x(n)\right)^{-\frac{b_2}{b_1+b_2}} = \frac{n}{\zeta_x(n)}.$$
We know that the saddle point approach to dominant singularity as $n\to\infty$, let's rewrite the above equation as
$$\frac{1}{(1-(b_1+b_2)x\zeta_x(n))^{\frac{b_2}{b_1+b_2}}}\Big((1-(b_1+b_2)x\zeta_x(n))^{-\frac{b_1}{b_1+b_2}} + a_0\Big) = \frac{n}{\zeta_x(n)}.$$
So asymptotically,
$$1-(b_1+b_2)x\,\zeta_x(n) \sim \left(\frac{a_0}{(b_1+b_2)x\,n}\right)^{\frac{b_1+b_2}{b_2}}.$$
So
$$\zeta_x(n) = \frac{1}{(b_1+b_2)x}\left(1-\left(\frac{a_0}{(b_1+b_2)x\,n}\right)^{\frac{b_1+b_2}{b_2}} + o\Big(n^{-\frac{b_1+b_2}{b_2}}\Big)\right).$$
We can check that $F(z,x)$ is H-admissible, the coefficient extraction admits the classical saddle–point estimate
$$[t^n]F(t,x) \sim \frac{F(\zeta_x(n),x)}{\zeta_x(n)^n\sqrt{2\pi h''(\zeta_x(n),x)}}.$$
So one obtains
$$p_n(x) \sim \frac{F(\zeta_x(n),x)}{F(\zeta_1(n),1)}\Big(\frac{\zeta_1(n)}{\zeta_x(n)}\Big)^n K_n(x)(1+o(1)), \qquad K_n(x) = \sqrt{\frac{n+\zeta_1(n)^2h''(\zeta_1(n),1)}{n+\zeta_x(n)^2h''(\zeta_x(n),x)}}.$$
Now, I want to apply the Generalized quasi-powers, we need $p_n = \exp(h_n)$, so
$$h_n(x) = \log p_n(x) = h(\zeta_x(n),x) - h(\zeta_1(n),1) + n\log\frac{\zeta_1(n)}{\zeta_x(n)} + \log K_n(x) + o(1).$$
Checking carefully, on has
$$\frac{h_n'''(x)}{\big(h_n'(1)+h_n''(1)\big)^{3/2}} \to 0 \quad (n\to\infty).$$
So the Generalized quasi-powers theorem implies that the random variable associated X_n converges in distribution to a normal law with mean $h_n'(1)$ and variance $h_n'(1)+h_n''(1)$.

..... COMPUTATION TO BE CHECK CAREFULLY AND CONTINUED... SOME CONDITIONS REQUIRED!!!

2.6 Case 3A: $b_1, a_2, b_1+b_2 \neq 0$

$$F(t,x) = \sum_{n\geq 0}\sum_{k=0}^{n} T(n,k)x^k\frac{t^n}{n!} = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$
$$\times\left(1-\frac{a_2}{b_1x}\Big[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\Big]\right)^{-\frac{a_0+a_2}{a_2}}.$$

Proposition 2.4. *Let $(T(n,k))$ be sequence satisfying the usual initial condition* (1) *and*
$$T(n,k) = (a_2n - a_2k + a_0)\,T(n-1,k) + (b_2n + b_1k + b_0)\,T(n-1,k-1).$$
Assume that all coefficients are non-negative and one of the following conditions holds

(1) $a_2 > b1$ *and* $\left(1-\frac{b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}} < 2$ *and* $b_2 \neq 0$.

(2) Second item

$F(t,x)$ be the exponential bivariate generating function. Then The random variable X_n of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a a Gaussian with speed of convergence $O(n^{-\frac{1}{2}})$, mean and variance

$$\mu_n \sim \left(1 - \frac{b_1+b_2}{a_2} \cdot \frac{\left(1-\frac{b_1}{a_2}\right)^{\frac{b_2}{b_1}}}{1-A}\right) n, \quad \sigma_n^2 \sim \left[\frac{(b_1+b_2)^2}{a_2^2} \cdot \frac{\left(1-\frac{b_1}{a_2}\right)^{\frac{b_2}{b_1}-1}}{(1-A)^2} \cdot \left[\frac{a_2}{b_1+b_2}(A-1)+1\right]\right] n,$$

where $A = \left(1-\frac{b_1}{a_2}\right)^{\frac{b_2}{b_1}+1}$.

Proof. We notice first that the non-negativity of all $T(n,k)$ implies

$$a_2, \quad a_2+a_0, \quad b_2, \quad b_1+b_2, \quad b_1+b_2+b_0 \geq 0. \tag{29}$$

Note that the singularity at $x=0$ is removable. We want to use Algebraic singularity schema, meanly we try to write

$$F(t,x) = A(t,x) + B(t,x)C(t,x)^{-\alpha}. \tag{30}$$

$A = 0, B(t,x) = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$ and $C(t,x) = 1 - \frac{a_2}{b_1 x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]$. for good choice of $r < \frac{1}{b_1+b_2}$. Under some conditions on the sign a_i, b_i, non-degeneracy is also verified. Let

$$\rho(x) = \frac{1}{(b_1+b_2)x}\left(1-\left(1-\frac{b_1}{a_2}x\right)^{\frac{b_1+b_2}{b_1}}\right),$$

whose the singularity at $x=0$ is removable.
The condition (1) implies $\rho = \rho(1)$ is dominant singularity comparing to the singularity coming from $B(t,x)$ that is $s = \frac{1}{b_1+b_2}$. Also for good choice of $r \in (\rho, s)$, the root of $C(t,1)$ is simple and unique in $|t| \leq r$. So the analytic perturbation condition is verified. Moreover, one can compute is

$$\partial_t C(\rho,1) = -a_2\left(1-\frac{b_1}{a_2}\right)^{-\frac{b_2}{b_1}}, \quad \partial_x C(\rho,1) = 1 - \frac{a_2}{b_1+b_2}\left(1-\left(1-\frac{b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}\right)\left(1-\frac{b_1}{a_2}\right)^{-b_2/b_1}.$$

It is clear that this $\partial_t C(\rho,1) \neq 0$. To see $\partial_x C(\rho,1) \neq 0$, we use the fact $b_1+b_2 \neq 0$, so $\frac{b_2}{a_2} \neq -\frac{b_1}{a_2}$. With the injectivity of $y \mapsto \frac{\log(1+y)}{y}$, we can conclude then

$$\frac{\log(1+\frac{b_2}{a_2})}{\frac{b_2}{a_2}} \neq \frac{\log(1-\frac{b_1}{a_2})}{-\frac{b_1}{a_2}} \implies \left(1-\frac{b_1}{a_2}\right)^{b_2}\left(1+\frac{b_2}{a_2}\right)^{b_1} \neq 1 \implies \partial_x C(\rho,1) \neq 0.$$

So we have the non-degeneracy condition, with $\rho(x)$ is root of $C(t,x)$ near to $x=1$.
Next, we see that

$$\mathfrak{v}\left(\frac{\rho(1)}{\rho(x)}\right) = \frac{(b_1+b_2)^2}{a_2^2} \cdot \frac{\left(1-\frac{b_1}{a_2}\right)^{\frac{b_2}{b_1}-1}}{\left(1-\left(1-\frac{b_1}{a_2}\right)^{\frac{b_2}{b_1}+1}\right)^2} \cdot \left[\frac{a_2}{b_1+b_2}\left(\left(1-\frac{b_1}{a_2}\right)^{\frac{b_2}{b_1}+1}-1\right)+1\right].$$

To see that it is always nonzero. We notice first that the only nontrivial is on the last part that is zero if and only if

$$\left(1-\frac{b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}} = 1-\frac{b_1+b_2}{a_2} = 1-\frac{b_1}{a_2}\times\frac{b_1+b_2}{b_1}.$$

It is impossible by the Bernoulli's inequality because $-\frac{b_1}{a_2} > -1$ and $\frac{b_1+b_2}{b_1} > 1$ (as $b_2 > 0$). Thus the variability is also verified. Therefore we can apply Theorem 2.3 to conclude that the random variable X_n of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a Gaussian with speed of convergence that is $O(n^{-\frac{1}{2}})$. The mean μ_n and the standard deviation σ_n^2 are asymptotically linear in n. More precisely $\mu_n \sim \mathfrak{m}\left(\frac{\rho(1)}{\rho(x)}\right)n$ and and $\sigma_n^2 \sim \mathfrak{v}\left(\frac{\rho(1)}{\rho(u)}\right)n$. Noting $\mathfrak{m}\left(\frac{\rho(1)}{\rho(x)}\right) = -\frac{\rho'(1)}{\rho(1)}$, we have explicitly μ_n and σ_n^2. $\square$

n	$E(X_n)$	μ_n	Relative error	$V(X_n)$	σ_n^2	Relative error
50	33.166099	33.589744	0.01277	15.381102	16.326101	0.06144
100	66.795100	67.179487	0.00575	31.718701	32.652202	0.02943
200	134.007185	134.358974	0.00263	64.413255	65.304405	0.01383
500	335.570576	335.897436	0.00097	162.412355	163.261012	0.00523
1000	671.477328	671.794872	0.00047	325.690874	326.522025	0.00255
1200	805.837907	806.153846	0.00039	390.998370	391.826430	0.00212
1500	1007.377988	1007.692308	0.00031	488.958120	489.783038	0.00169
1900	1276.097311	1276.410256	0.00025	619.569616	620.391848	0.00133
2000	1343.277058	1343.589744	0.00023	652.222326	653.044050	0.00126
2500	1679.175481	1679.487180	0.00019	815.485279	816.305062	0.00101

Table 3: Comparison of exact and approximate values for $a_0 = 8, b_0 = 10, b_2 = 5, b_1 = -1, a_2 = 2$.

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